SYNTRIASS

AIRON AGENT PASSPORT

AI agents need passports before they enter regulated infrastructure.

AIRON combines a portable Agent Passport layer with kernel-level AEGIS enforcement so autonomous agents can be identified, constrained, verified, revoked, and audited.

AIP: passport and delegation

AEGIS: eBPF/XDP enforcement

TRL-5 pilot-ready path



PASS / TOXIC

Offline and online passport verification result

CI-verified

AIP service, SDK, and verifier run in GitHub Actions

TRL-4 -> 5

Lab validation complete; relevant-environment pilots are next

PROBLEM

AI agents are being deployed without an accountability boundary.

In regulated workflows, a model output is not enough. Auditors need to know which agent acted, who authorized it, what it was allowed to do, and whether that authority was still valid.

Anonymous automation

Shared API keys and service accounts hide the true actor.

->

Scope creep

Sub-agents inherit tools without mathematically bounded authority.

->

No court-grade trail

Logs are fragmented and cannot prove delegation lineage.

->

Liability gap

A breach can scale at machine speed while ownership stays unclear.

SOLUTION

AIRON Agent Passport gives every agent a portable credential.

The passport is a signed JSON document that travels with the agent and can be verified before tool execution or during offline audit.

IDENTITY

Agent ID, friendly name, and public key fields.

AUTHORITY

Issuer identity and signed lineage bind the passport to a parent AIP token.

```
{
  header: AIP-PASSPORT,
  subject: agent + keys,
  issuer: AIRON root,
  constraints: scopes + expiry,
  lineage: signed authority,
  environmental_lock: runtime,
  proof: signature
}
```

CONSTRAINTS

Capability list, expiry, and max delegation depth define what the agent can do.

PROOF

Ed25519 proof in Phase 1; ML-DSA field reserved for hybrid PQC upgrade.

Verify online | Verify offline | Tamper = TOXIC

PRODUCT

Passporting an agent is now an executable workflow.

The current repo exposes issuance, verification, tamper rejection, offline audit, and Python SDK UX.

1. Issue AIP token

POST /v1/identity/issue creates a signed agent token.

->

2. Issue passport

POST /v1/passports/issue validates the token and signs the passport body.

->

3. Verify before tool

POST /v1/passports/verify returns PASS for valid authority.

->

4. Tamper test

Capability mutation returns TOXIC instead of silent acceptance.

Python: AgentPassport(config).issue(...)

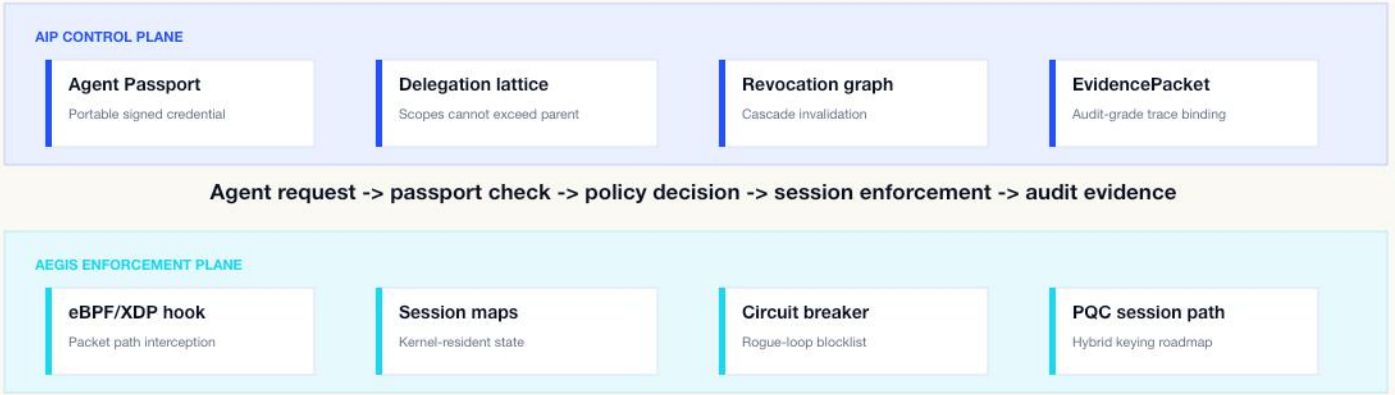
CLI: airon-passport verify agent.passport.json

Developer outcome: one credential can be passed across LangChain agents, API tools, notebooks, and audit workflows.

PLATFORM

AIP governs the actor; AEGIS enforces the boundary.

Together they create a trust gateway for regulated agent traffic: identity decisions in user space, fast containment in kernel space.



VALIDATION

The story has moved from code fragments to repeatable evidence.

AIP now has a testable passport schema, service endpoints, offline verifier, Python SDK, and Colab pilot harness.

113
aip-core tests in local validation

25
aip-service tests including passport verifier checks

7
Python SDK tests covering passport UX

PASS
GitHub AIP Validation for passport implementation

CURRENT TRL

TRL-4

Pilot evidence is next.

EVIDENCE LADDER

- Schema: AgentPassport v1 fields and validation helpers
- Service: /v1/passports/issue and /verify
- Verifier: airon-passport offline PASS / TOXIC audit
- Pilot: Colab notebook exports pilot_results.json

MARKET WEDGE

Regulated AI needs a trust gateway before pilots become production.

AIRON starts where the cost of anonymous autonomous action is highest: financial workflows, critical operations, and defense autonomy.

WEDGE 1

BFSI and fintech

AI agents reading invoices, reconciling ledgers, moving cases through payment operations.

Trigger: auditability before agentic automation scales.

WEDGE 2

Enterprise AI teams

LangChain and notebook-based agents that need portable credentials and tamper-evident tool access.

Trigger: internal controls for autonomous workflows.

WEDGE 3

Defense / critical infra

Autonomous systems requiring a black-box record of authority, runtime, and allowed action.

Trigger: mission accountability and fail-closed behavior.

Pilot wedge

Start with Agent Passport SDK and Colab evidence. Expand to AIRON Trust Gateway once workflow authority is proven.

MARKET SIZE

Agent identity sits inside a fast-growing trust infrastructure market.

AIRON is sized as a new trust layer across identity, AI governance, and agent infrastructure. TAM uses public market reports; SAM and SOM are AIRON operating assumptions.

TAM

~$38B

Global adjacent 2026 spend pool: IAM, AI agents, and AI governance.

SAM

~$3.8B

Regulated agent trust layer: BFSI, enterprise AI, government, and defense.

SOM

$3-5M ARR

24-month wedge: paid pilots, early customers, and defense design partners.

Source basis: IAM ~$26.77B 2025 baseline, AI agents ~$10.91B in 2026, AI governance spend ~$0.49B in 2026. SAM assumes a 10% regulated trust-infrastructure slice; SOM assumes 20-30 paid pilot/customer conversions.

BUSINESS MODEL

Monetize agent trust as identity, audit, and usage infrastructure.

Open-source SDK adoption creates developer pull; regulated teams convert to paid identity, audit, and enforcement services.

MODEL 1

Identity-as-a-Service

Passport issuance, registry, revocation, scoped delegation.

Per agent / month + API calls

MODEL 2

Audit-as-a-Service

EvidencePacket storage, export, verifier reports, retention.

Per audit packet + retention tier

MODEL 3

Usage-based pricing

Issue, verify, revoke, and evidence API metering.

Per 1K verifications/actions

MODEL 4

Enterprise / defense

Private control plane, support, air-gapped deployment, AEGIS nodes.

Annual license + capacity

Commercial path: free SDK -> paid pilots -> Identity-as-a-Service / Audit-as-a-Service -> enterprise and defense deployments.

ROADMAP

Funding converts lab truth into revenue, IP, and defense-ready evidence.

The next gate is controlled pilot evidence plus business proof: design partners, paying customers, global IP, and defense validation.



NOW

TRL-4

Repo + CI validation, SDK, Colab harness, pilot pack

30 DAYS

Design partners

BFSI, enterprise AI, and defense pilot conversations

90 DAYS

Revenue + IP

Paid pilot LOIs, global IP filings, PQC hardening

180 DAYS

Defense GTM

Reference customers, defense demo, AEGIS lab dossier

Use of funds: pilot deployment, revenue conversion, global IP, ML-DSA hardening, defense demos, and bare-metal AEGIS test infrastructure.

USP

AIRON is the passport layer for agents, not just IAM, workload ID, or observability.

The wedge is not replacing established identity stacks. The wedge is the missing portable credential that proves an agent's authority, scope, lineage, and revocation state at the point of action.

Okta / IAM

Govern users, apps, NHIs, and enterprise access.

Adds passport proof.

SPIFFE / SPIRE

Issue workload identities and SVIDs for services.

Adds authority lineage.

Maxim AI / evals

Evaluate, simulate, and observe agent behavior.

Adds verified authority.

API gateways

Route APIs and enforce request-level access.

Adds audit evidence.

AIRON DIFFERENCE

Portable credential, cryptographic finality, enforceable boundary.

- Travels with LangChain, notebooks, APIs, and audits.
- Binds human authority to agent action.
- Limits scope with lattice, expiry, and revocation.
- Verifies offline as PASS or TOXIC.
- Connects to AEGIS for network containment.

Positioning: works beside Okta and SPIFFE; competes where agent-specific proof, revocation, and forensic accountability are mandatory.

ASK

Back the Agent Passport pilot and the AIRON Trust Gateway roadmap.



We are seeking VC conversations, RDI/TDB support, and design partners: agent workflows first, gateway infrastructure next.

1	VC investors: Lead or join a strategic pre-seed/seed round for Agent Passport pilots and GTM.
2	Design partners: Run the Colab pilot against a deployed AIP service and return pilot_results.json.
3	RDI support: Fund TRL-5/6 validation: deployment, cryptographic hardening, and bare-metal lab.
4	Technical diligence: Review the repo, CI artifacts, passport verifier, and validation ladder.

Next step

Send the deck to VCs with the pilot ask: capital, regulated intros, and technical diligence on the Agent Passport evidence stack.